

SOC OPERATIONAL CHEAT SHEET

SECURITY OPERATIONS CENTER // DEFENSIVE CYBER PROTOCOLS

STATUS: ACTIVE 24/7/365
LEVEL: DEFENSIVE baseline

01. SOC BASELINE & CORE MISSION

A **SOC (Security Operations Center)** is a dedicated 24/7 facility operated by a specialized team to continuously monitor an organization's network, assets, and systems. The ultimate goal is to identify suspicious activity and prevent damage to critical digital assets (secrets).

DETECTION CAPABILITIES

- **Vulnerabilities:** System/OS software weaknesses requiring patches.
- **Unauthorized Activity:** Compromised credentials, unusual geo-logins.
- **Policy Violations:** Downloading pirated software, insecure data transfers.
- **Intrusions:** Web app exploits, malware infections, unauthorized access.

RESPONSE CAPABILITIES

- **Incident Containment:** Minimizing threat impact immediately upon detection.
- **Root Cause Analysis:** Investigating origin vectors to prevent recurrence.
- **Eradication & Recovery:** Removing threats and restoring standard operations.
- **Support:** Assisting dedicated Incident Response (IR) teams.

02. THE 3 PILLARS OF A MATURE SOC

PEOPLE

CRITICAL

Human operators required to analyze noise, eliminate false positives, and execute critical decisions.

PROCESS

STRUCTURE

Standardized workflows (Triage, Escalation, Forensics) ensuring uniform, timely responses.

TECHNOLOGY

ENABLER

Security solutions (SIEM, EDR, Firewall) centralizing log data and automating alerts.

03. SOC TEAM STRUCTURE & ROLES

SOC Analyst (Level 1)

Triage

First responders to security alerts. Perform initial alert triage, assess basic severity, and filter noise/false positives.

SOC Analyst (Level 2)

Investigation

Drives deeper investigations. Correlates data from multiple sources to confirm complex incidents and coordinate response.

SOC Analyst (Level 3)

Threat Hunting

Experienced specialists. Proactively hunt for hidden threat indicators and lead severe incident containment & recovery.

Security Engineer

Infrastructure

Deploys, integrates, and maintains security tools and infrastructure to ensure constant operational uptime.

Detection Engineer

Rules & Logic

Builds, tunes, and optimizes detection rules and logic behind security tools to spot emerging attacker techniques.

SOC Manager

Management

Oversees daily operations, manages workflow processes, and liaises directly with the CISO regarding threat posture.

■ 04. SOC PROCESSES & THE 5 WS FRAMEWORK

Alert Triage is the cornerstone process in a SOC. When an alert triggers, analysts utilize the 5 Ws framework to systematically evaluate, categorize, and prioritize the event.

THE 5 WS	OPERATIONAL QUESTION	INCIDENT CASE STUDY EXAMPLE (MALWARE DETECTION)
WHAT?	What type of threat or activity was detected?	Malicious file executable detected inside the company network.
WHEN?	When did the security event take place?	Timestamp: 13:20 on June 5, 2024.
WHERE?	Where is the host/asset located?	Host workstation directory: GEORGE PC .
WHO?	Who was logged in or triggered the action?	Executed under account context: George .
WHY?	Why did this occur (root cause)?	Downloaded from pirated software site (policy violation for unapproved software).

■ 05. ESCALATION, REPORTING & FORENSICS

REPORTING & ESCALATION

- **Ticket Creation:** Confirmed alerts are escalated as structured tickets to L2/L3 analysts.
- **Documentation:** Reports must detail all 5 Ws alongside analyst notes.
- **Evidence Artifacts:** Include screenshots, log snippets, and file hashes as evidence.

DFIR & ROOT CAUSE ANALYSIS

- **Incident Response:** Triggered for critical severities to contain malware or breach attempts.
- **Forensic Artifacts:** Deep analysis of system memory, disk, and network packets.
- **Root Cause Identification:** Pinpointing exact entry vectors to close security gaps.

■ 06. SECURITY TECHNOLOGY STACK

SIEM – Security Information & Event Management

Centralized platform that ingests logs from diverse network sources. Employs detection rules, machine learning, user behavior analytics (UEBA), and threat intelligence correlation.

Key Note: SIEM primarily provides *Detection* capabilities in the SOC.

EDR – Endpoint Detection & Response

Monitors endpoints (workstations, servers) in real time. Offers deep historical visibility into process trees and file executions, allowing rapid containment/isolation with a few clicks.

FIREWALL

Acts as a perimeter defense between internal network zones and external networks (Internet). Filters traffic using predefined rules to block unauthorized connections.

ADDITIONAL SOC TOOLS

SOAR: Security Orchestration, Automation & Response | **XDR:** Extended Detection & Response | **IDS/IPS:** Intrusion Detection/Prevention Systems | **EPP:** Endpoint Protection Platform.